SETISIN INJECTION

DESCRIPTION:SETISIN INJECTION (1ML AMP) is a clear, almost colourless solution.

COMPOSITION: Each ampoule contains Neostigmine Methylsulphate BP 2.5 mg/ml.

PHARMACODYNAMICS: Cholinesterase inhibitor which reversibly inhibits the hydrolysis of acetylcholine thereby potentiating its action. Neostigmine is a anticholinesterase agent which inhibits reversibly the hydrolysis of acetylcholine. As a result acetylcholine accumulates at cholinergic synapses and its effects are prolonged and exaggerated. Neostigmine is therefore capable of producing a generalised cholinergic response, including miosis, increased tonus of intestinal and skeletal musculature, constriction of bronchi and ureters, bradycardia and stimulation of salivary and sweat glands. In addition neostigmine has a direct cholinomimetic effect on skeletal muscle and to a lesser extent to increase the activity of smooth muscle.

PHARMACOKINETICS: Because of its quaternary ammonium structure, in moderate doses, neostigmine does not cross the blood-brain barrier to produce CNS effects. Extremely high doses, however, produce CNS stimulation followed by CNS depression.
Following IV administration the elimination half-life ranges from 47 to 60 minutes and after IM administration 50 to 91 minutes. Approximately 80% of a single IM dose of neostigmine is excreted in the urine in 24 hours, about 50% as unchanged drug and the remainder as metabolites. The major site of uptake is in the liver. It is metabolised partly by the hydrolysis of the ester linkage and partly by microsomal enzymes in the liver.

INDICATIONS: Reversal of the effects on nondepolarising neuromuscular blocking agents (e.g. tubocurarine, pancuronium, etc.). Prophylaxis and treatment of post-operative intestinal atony and urinary retention. Also treatment of myasthenia gravis.

RECOMMENDED DOSAGE: The following doses are approximately equivalent in effect: 0.5 mg/IV= 1.0 - 1.5 mg IM or SC.
Antagonist to Nondepolarising Neuromuscular Blockade: Usually, reversal of neuromuscular blockade with Neostigmine Injection BP should not be attempted until spontaneous recovery from paralysis is evident. It is recommended that the patient be well ventilated and patent airway maintained until complete recovery of normal respiration is affirmed.
Adult: A single dose of neostigmine 2 to 3 mg with atropine sulphate 0.6-1.2 mg by slow IV injection over 1 minute. The recommended ratio of atropine to neostigmine is 1:2 to 1:3. The maximum recommended dose of neostigmine in adult is 5 mg.
Children: The suggested dose in children is 0.05 mg/kg/dose and atropine sulphate 0.02 mg/kg/dose by slow IV injection over 1 minute. Maximum recommended dose of neostigmine in children is 2.5 mg. The two drugs are often given simultaneously, but in patients with bradycardia, the pulse rate should be increased to about 80 beats/minute with atropine before administering neostigmine. The speed of recovery from neuromuscular blockade is primarily determined by the intensity of the block at the time of antagonism. It Is also influenced by other factors including the presence of drugs (eg. anaesthetic drugs, antibiotics and antiarrhythmic drugs) and physiological changes (eg. electrolyte and acid-base imbalance, renal impairment) The factors may prevent successful reversal with Neostigmine Injection BP or lead to re-curarisation after apparently successful reversal. It is imperative that the patients should not be left unattended until these possibilities have been excluded.
Myasthenia Gravis: 1 mg to 2.5 mg given I.M. or S.C. at intervals throughout the day when greater strength may be needed (eg. morning and before meals) giving a total daily dose of 5 to 20 mg. Duration of action of a single dose is 2 to 4 hours.
Neonates: 0.05 – 0.25 mg IM every 2-4 hours, half an hour before feeding. Treatment is not usually required beyond 8 weeks of age. Because the condition is usually self-limiting the daily dosage should gradually be reduced until the drug can be withdrawn.
Older Children: 0.2 to 0.5 mg by injection as required. Dosage should be adjusted according to response.
When large doses of neostigmine are given to myasthenic patients, atropine sulphate may be required to counteract the muscarinic side effects.
Intestinal Atony: Prophylaxis. 0.25 mg SC or IM before or immediately after the operation, repeated every 4 to 6 hours, for 2 to 3 days. Treatment: 0.5 mg SC, IM or IV repeated at intervals of 4 to 5 hours.
Urinary Retention: Prophylaxis. 0.25 mg as for intestinal atony.
Treatment: 0.5 mg SC or IM and apply heat to lower abdomen. After patient has voided continue 0.5 mg SC or IM every 3 hours for at least 5 injections. If there has been no urinary response within one hour of the first dose, the patient should be catheterised.

ROUTE OF ADMINISTRATION: For i.v., i.m. and s.c. injection

CONTRAINDICATIONS: Mechanical obstruction of intestinal or urinary tract. Known hypersensitivity to neostigmine.

WARNING & PRECAUTIONS: Use with caution in patients with bronchial asthma, cardiac disease, epilepsy, hypotension, parkinsonism or peptic ulceration.
With large doses, simultaneous parenteral administration of atropine sulphate may be advisable. Atropine sulphate should always be available along with other anti-shock medications (adrenaline) in case of hypertensitivity to neostigmine.
Neostigmine may prolong the depolarising neuromuscular blocking agent of suxamethonium and prolonged apnoea may result (see Drug Interactions). Neostigmine should not be given whilst anaesthesia with cyclopropane and halothane continues but may be used after withdrawal of these agents.

INTERACTION WITH OTHER MEDICATIONS: Corticosteroids may decrease the anticholinesterase effects of neostigmine. Conversely anticholinesterase effects may increase after stopping corticosteroids. Neostigmine may prolong the Phase 1 block of depolarising muscle relaxants such as suxamethonium. Prolonged respiratory depression with extended periods of apnoea may occur. Atropine reverses the muscarinic effects of neostigmine. Atropine reverses the muscarinic effects of neostigmine. This interaction is used to counteract the muscarinic symptoms of neostigmine toxicity, however masking the signs of overdosage can lead to inadvertent induction of cholinergic crisis with the use of atropine. Anticholinesterase agents can be effective in reversing neuromuscular block induced by aminoglycoside antibiotics. Aminoglycoside antibiotics, local and some general anaesthetics, antiarrhythmic agents and other drugs that interfere with neuromuscular transmission should be used cautiously, if at all, in patients with myasthenia gravis. The dose of neostigmine may need to be increased accordingly.

USE IN PREGNANCY & LACTATION: Use in pregnancy: Neostigmine. The safety of neostigmine in pregnancy has not been established with respect to possible adverse effects on foetal development. Anticholinesterase agents may cause uterine irritability and induce premature labour when given IV to pregnant women near term. Therefore neostigmine should not be used in pregnant women or those likely to become pregnant unless the expected benefits outweigh any potential risk.
Use in lactation: Evidence indicates that only negligible amounts of neostigmine enter the breast milk, nevertheless the possibility of adverse effects on the breast feeding infant should be considered.

SIDE EFFECTS: Adverse reactions are generally associated with neostigmine overdosage are:
Cardiovascular: Cardiac arrhythmias (especially bradycardia), hypotension, cardiac arrest.
Central Nervous System: Headache, convulsions, coma, restlessness, slurred speech, agitation and fear.
Gastrointestinal: Nausea, vomiting, diarrhoea, abdominal cramps, increased peristalsis and involuntary defaecation.
Genitourinaly: Involuntary urination or desire to urinate.
Musculoskeletal: Muscle cramps, fasciculation, general weakness and paralysis.
Respiratory: Increased bronchial secretion, bronchospasm, respiratory depression, tight chest and wheezing.
Allergic: allergic reactions including anaphylaxis.
Skin: rash and urticaria.
Other: Increased sweating and salivation, miosis, vision changes, nystagmus and lacrimation.

SYMPTOMS AND TREATMENT OF OVERDOSE: Nicotinic and muscarinic effects (see *Adverse Effects*). Discontinue all cholinergic medication. Muscarinic effects are controlled with IV atropine sulphate (1 mg to 2 mg) followed by IM atropine sulphate every 2 to 4 hours. Assistance of ventilation may be required.

INCOMPATIBILITIES: Neostigmine methylsulphate and atropine must not be injected with the same syringe.

STORAGE CONDITIONS: Store below 30ºC. Protect from light. KEEP OUT OF REACH OF CHILDREN. *JAUHKAN DARIPADA KANAK-KANAK.*

PACK SIZE: Box of 10's in 1 ml amber ampoules.

SHELF LIFE: Please refer to the outer package

PRODUCT REGISTRATION HOLDER & MANUFACTURER:
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